

JORDAN BANH

+1(613) 716-7146 ◇ Ottawa, ON

jordanbanh@cmail.carleton.ca ◇ [linkedin.com/in/jordan-banh](https://www.linkedin.com/in/jordan-banh)

EDUCATION

Bachelor of Engineering, Biomedical and Mechanical Engineering

Expected 2028

Carleton University, Ottawa ON

Coursework: Thermodynamics, Fluid Mechanics, Dynamics, Mechanics of Solids, Engineering Materials

EXPERIENCE

Maintenance & Reliability Engineering Co-op

Jun 2025 – Aug 2026

CertainTeed Insulation Canada (Saint-Gobain)

Ottawa, ON

- Evaluated a replacement chopper frame that cut peak stress 10.5x in the side plane and 6.5x frontally, resolving fatigue cracking from repeated weld repairs; verified dimensional fit on site.
- Reselected material for a 24 in curing oven fan running near 1600°F, comparing mild steel, stainless, and Corten on cost and oxidation resistance.
- Analyzed rated lift capacity of a Toyota LP forklift at an extended load center to confirm safe handling limits.
- Performed vibration analysis and root cause diagnostics on rotating equipment, including an intermittent star-delta contactor fault on an Atlas Copco ZT160 compressor.
- Modeled guarding and equipment geometry in Fusion 360 and AutoCAD LT; produced a 1:1 scale P&ID of the undocumented plant compressed air network.
- Authored CAPEX justifications, SOPs, SMPs, and LOTO procedures to TSSA requirements; managed work orders in SAP.

Coding and Robotics Instructor

Mar 2024 – Sept 2024

Resolute Center

Ottawa, ON

- Taught Python, C, and LEGO SPIKE robotics to over 60 students aged 7–15.

PROJECTS

Compressor Centralization and SCADA Integration

Sept 2025 – Jun 2026

- Traced excess air production to a decentralized compressed air system; planned Ignition SCADA integration for a TURBO-AIR 2000 centrifugal compressor to enable automated load balancing.
- Specified IO-Link flow instrumentation on a 6 in header, coordinated vendor quotes, and built the business case projecting \$250K in annual electricity savings.

Reliability Data and Machine Documentation Tooling

2026

- Built Excel VBA tools computing MTBF and MTTR from breakdown logs and a Python OCR pipeline (OpenCV, openpyxl) replacing manual fault-count transcription from machine HMIs; completed a WCM Yellow Belt machine study on the Strapper lines.

Roof Truss Design (Academic)

Apr 2024 – May 2024

- Performed member force analysis to size a roof truss for load distribution; modeled in Fusion 360.

SKILLS

Engineering	Stress and fatigue evaluation, material selection, vibration analysis, root cause analysis, MTBF/MTTR reliability, P&ID development, LOTO and TSSA compliance
Software	Fusion 360, AutoCAD LT, SAP PM, Ignition SCADA, Excel (advanced)
Programming	Python (OpenCV, openpyxl, pandas), VBA, C, JavaScript/React